

SNEHA GHOSH

AI Developer — GenAI Engineer

📍 Kolkata, West Bengal, India ✉ snehanew2003@gmail.com ☎ (+91) 9832178973 📁 Portfolio in LinkedIn 🐙 GitHub

OVERVIEW

AI/ML Engineer and B.Tech graduate in Information Technology with hands-on experience building production-oriented intelligent systems across NLP, Computer Vision, RAG, Voice AI, and document intelligence. Skilled in developing end-to-end AI solutions using Python, FastAPI, TensorFlow, LangChain, PostgreSQL, and modern AI frameworks. Strong foundation in machine learning, deep learning, and data-driven problem solving, with experience transforming real-world business requirements into scalable AI applications.

EDUCATION

College of Engineering and Management, Kolaghat Bachelor of Technology in Information Technology — CGPA: 8.25	2021 – 2025
Kendriya Vidyalaya Ranaghat Class XII, Science — 87.4%	2020 – 2021
Kendriya Vidyalaya Ranaghat Class X — 85.8%	2018 – 2019

SKILLS

Technical Skills	Python, SQL, Statistics, Probability, Linear Algebra, Calculus, Hypothesis Testing
AI/ML & Deep Learning	TensorFlow, Keras, Scikit-Learn, PyTorch, Pandas, NumPy, Matplotlib, Scipy, Seaborn, NLP, Computer Vision
Backend & AI Systems	FastAPI, LangChain, LangGraph, RAG, Vector Databases, PostgreSQL, Docker, CrewAI, MCP, n8n
Tools & Platforms	Power BI, Excel, Streamlit, Git
Soft Skills	Adaptability, Growth Mindset, Open to Feedback, Communication Skills, Team Player

EXPERIENCE

AI/ML Intern <i>Darpin Dynamics</i>	December 2025 – April 2026
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AI Voice Calling Automation System

- Built an end-to-end voice AI pipeline for automated 24/7 customer call handling, enabling real-time conversational responses without human intervention.
- Integrated **PostgreSQL** as the backend data store for persisting call logs, customer records, and session states.
- Developed a companion analytics dashboard monitoring sentiment trends, multilingual transcripts, and customer interaction quality for business intelligence.

Tech Stack: Voice AI Platform, PostgreSQL, Python

Multilingual Feedback AI System

- Built a feedback ingestion system accepting customer input in any Indian language, performing automatic translation and transcription using speech and NLP models across 10+ regional languages.
- Stored structured feedback with sentiment tagging, enabling downstream analysis across regional language inputs.
- Designed an automated feedback categorization system grouping user inputs into actionable categories, reducing manual triage effort for business teams.

Tech Stack: Python, NLP, Speech Models, PostgreSQL

AI Memory Assistant for Gmail & Calendar

- Designed a personal memory module syncing Gmail and Google Calendar via OAuth, storing indexed content through a local persistence layer.
- Implemented a **RAG** pipeline allowing users to query their email and calendar history in natural language.
- Enabled AI-generated SQL queries from natural language prompts for structured retrieval from synced data stored in **PostgreSQL**.

Tech Stack: Python, RAG, LangChain, PostgreSQL, OAuth

Forensic Face Recognition & Restoration System

- Developed a forensic face recognition pipeline using **ResNet** for deep feature extraction and cosine similarity scoring, designed to identify individuals from degraded or post-mortem imagery with preprocessing for noise reduction and contrast normalization.

- Assisted in building a diffusion-model-based face restoration module using GFPGAN to reconstruct damaged facial images as a high-quality pre-step to recognition.

Tech Stack: Python, ResNet, GFPGAN, OpenCV

Medical Bill PDF-to-JSON Extraction System

- Built a multi-stage intelligent pipeline to extract structured JSON from scanned and digital hospital bill PDFs using PaddleOCR and pdfplumber, achieving high accuracy on complex multi-page formats.
- Integrated vision-language models via Ollama (LLaVA, Qwen2.5) with a custom legibility scoring module to auto-route between OCR and direct extraction paths.
- Exposed the pipeline as a FastAPI REST endpoint accepting PDF uploads and returning fully validated JSON; included deskewing, denoising, and category-aware batching.

Tech Stack: Python, PaddleOCR, pdfplumber, Ollama, FastAPI, Docker

PROJECTS

Git Analyzer — *Multi-Agent System*

[GitHub](#)

- Built a two-phase RAG ingestion pipeline: Phase 1 summarizes git commit diffs using AST-aware before/after code context via an LLM; Phase 2 chunks source files at Python AST boundaries, markdown headers, and line windows — embeddings stored in ChromaDB.
- Implemented a 3-agent CrewAI workflow (Retriever → Reasoner → Responder) with 6-category intent classification to route natural language queries across architecture, history, file explanation, and dependency questions.
- Exposed FastAPI endpoints for background repo ingestion with live progress tracking, chat sessions, and full message history persisted in PostgreSQL.

Tech Stack: Python, FastAPI, CrewAI ChromaDB, PostgreSQL, GitPython

TravelMind AI — *LLM-Powered Travel Assistant*

[GitHub](#)

- Architected a full-stack AI travel assistant using FastAPI with LLM API integration, supporting multi-turn contextual conversations for destination planning, route optimization, and weather-aware recommendations.
- Designed a RAG-based QA module retrieving location-specific context from a vector store for accurate, grounded responses to complex travel queries.
- Built an interactive Streamlit frontend with dynamic session management, external API integrations (weather, geolocation), and real-time streaming responses.

Tech Stack: Python, FastAPI, LangChain, RAG, Vector DB, Streamlit

Autonomous AI Cyber Investigation Platform — *Multi-Agent Systems / Cybersecurity*

Ongoing

- Designing a multi-agent pipeline (Threat Intel → Log Correlation → Malware Analysis → Timeline Recon → Attack Graph) to autonomously investigate cybersecurity incidents end-to-end.
- Implementing a multi-agent debate mechanism where agents independently assess threat severity and attacker intent, then reconcile findings with confidence scoring.

Tech Stack: Python, CrewAI, FastAPI

Image Captioning System — *Multimodal Deep Learning*

[GitHub](#)

- Designed an encoder-decoder architecture using Xception CNN as a pretrained visual encoder and a custom LSTM decoder with attention-aware sequence generation, trained on the Flickr8k dataset.
- Implemented beam search decoding (beam width = 3), improving caption fluency and BLEU score performance over greedy decoding.
- Deployed the inference pipeline via Streamlit with real-time image upload, caption generation, and optimized preprocessing for low-latency response.

Tech Stack: Python, Xception, LSTM, Keras, Streamlit, Flickr8k

CERTIFICATIONS

Udemy: The Data Science Course – Complete Data Science Bootcamp (2025)

Covered data preprocessing, feature engineering, ML (regression, classification, clustering via Scikit-Learn), deep learning with TensorFlow/Keras, and statistical foundations on real-world datasets.

TCS iON: IIT Kharagpur AI4ICPS Certificate Programme

Industry AI workflows, key algorithms (SVM, KNN, Naïve Bayes), transformers, computer vision, hyperparameter tuning, and ethical AI deployment.

DASHBOARDS

Hospitality Domain Analysis (Power BI) — Revenue, occupancy, and booking trend analysis across hotel categories.

[GitHub](#)

Sales Insight Dashboard (Power BI) — Regional performance, product-level revenue trends, and YoY sales comparison.

[GitHub](#)